AARAMBH

FOLLOW-UP NAVIGATOR

Implementation Plan

Care That Fits Real Life

AI SYNERGY HACKATHON 2026

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1. Purpose and Implementation Objectives

This plan defines a practical route from the Aarambh Round 1 problem-solution brief to a demonstrable prototype and, separately, to a potential supervised pilot proposal. It retains the source brief's central goal: turn clinician-approved follow-up instructions into understandable and achievable next steps, surface barriers early, coordinate human action, and verify outcomes without automating clinical decisions.

1.1 Implementation objectives

- Build a traceable care-journey workflow from the approved instruction through understanding, barrier capture, staff action, record reconciliation, and verified outcome.
- Use validation and explicit status values to prevent missing, stale, contradictory, or unverified information from being silently treated as fact.
- Provide a patient-facing pathway and a staff-facing queue, with role-based permissions and a clear audit history.
- Keep any AI-generated explanation grounded in the original clinician-approved source and label it as AI-assisted.
- Provide travel/departure guidance and waiting-time forecasts only when underlying inputs pass defined validity checks.
- Validate the prototype against defined functional, privacy, usability, and safety acceptance tests before demonstrating readiness for any supervised pilot.

1.2 Intended result

The first release should be a demonstrator that proves the workflow with synthetic records. It should not claim clinical effectiveness, predictive performance, live hospital integration, or real-time queue accuracy unless those capabilities have actually been implemented and validated.

2. Problem Baseline and Design Constraints

The case brief describes a simulated high-volume district-hospital clinic. The signals below are treated as case-pack inputs for problem framing and test design, not audited statistics from a real institution.

Case-pack signal	Implementation consequence	Uncertainty to resolve
Approximately 900 outpatient visits/day in the simulated clinic	Do not rely on a staff member manually examining every record; provide a manageable, reason-labelled queue.	Actual clinic volumes, arrivals, staffing, and peak-hour patterns.
28% marked for follow-up did not return within the recommended window in a three-month sample	Build status reconciliation before treating a case as a missed visit.	Which records were clinically important, rescheduled, attended elsewhere, or documented incorrectly.
19% shared household phone numbers; 7% unreachable numbers	Implement contact preference/consent checks, neutral notifications, bounded retries, and staff-assisted alternatives.	Phone ownership, safe channel, consent, and accepted alternatives.
Instructions stored as variable free text	Preserve source text and version; validate fields; generate explanations only from approved information.	Required fields, approved glossaries, supported languages, and review policies.
Small records team	Use a transparent administrative task queue with owner, reason, due window, and next action.	Feasible workload, triage rules, ownership, and service levels.
No autonomous diagnosis, treatment changes, or emergency triage	Keep clinical judgement with authorised humans; enforce prohibited-action guardrails.	Local escalation procedure and approval roles.

2.1 Selected problem statement

Problem: Patients and hospital teams lack one reliable, accessible workflow to turn clinician-approved follow-up plans into understood and achievable next steps, identify barriers early, plan visits with realistic timing information, and verify care completion—without confusing poor records with non-attendance or automating clinical decisions.

2.2 Design hypotheses to validate

- Clear instructions plus a patient understanding check may be more useful than message delivery alone.
- Record reconciliation before outreach may reduce avoidable false missed-visit alerts.

- Waiting-time guidance may help only when data are current and validated, and must be separate from required appointment/check-in time.
- A rules-based, staff-reviewed workflow is the baseline for the initial prototype; machine-learning prioritisation is a later evaluation, not a prerequisite.

3. Scope, Exclusions, Assumptions, and Dependencies

3.1 In scope for the initial prototype

- Synthetic or properly de-identified patient/follow-up records with clinician-approved sample instructions.
- Input validation, source/version preservation, and explicit handling of missing/conflicting information.
- Patient-friendly explanation flow, an understanding check, and an assistance request path.
- Capture of contact preference, consent state, and patient-stated access barriers using minimum necessary data.
- Staff dashboard/queue, assignment, administrative action tracking, attendance reconciliation status, and outcome verification.
- Demonstration visit planning using labelled simulated inputs; forecasts suppressed when inputs are invalid.
- Role-based access, event/audit records, manual correction, and basic operational metrics.
- Automated and scripted acceptance tests corresponding to the source brief's AT-01 through AT-10.

3.2 Out of scope

- Autonomous diagnosis, prescribing, treatment modification, clinical urgency ranking, or emergency triage.
- Changing or overriding the clinician-approved instruction.
- Production use with identifiable real-patient data before institutional approval and privacy/security review.
- Claims of live hospital integration, live traffic tracking, or reliable real-time waiting estimates without dependable validated feeds.
- Enabling a no-show prediction model for patient-facing decisions before label definition, data-quality review, held-out evaluation, baseline comparison, and human-action protocol.
- Replacing the hospital's registration, electronic medical record, scheduling, or records system.

3.3 Planning assumptions

The source brief does not prescribe a programming language, database, hosting platform, team size, or delivery duration. The technical choices and week-by-week schedule below are therefore recommendations for planning and should be adapted to team capacity and available approvals.

3.4 Dependencies

- Approved sample follow-up fields and definitions of "due," "missed," "unverified," and "completed."
- An approved source/version convention for clinical instructions and a validated language glossary for any languages shown.
- Role definitions and rules for staff-assisted support, caregiver access, consent, and safe messaging.
- Synthetic sample records and expected outputs for acceptance testing.
- For timing demonstrations: appointment/check-in times, travel-duration input, buffer rules, and simulated or validated queue data, explicitly marked by provenance.

4. Proposed Operating Workflow

The system must support an end-to-end care journey rather than only a reminder. The following sequence is the core workflow for the first demonstrator.

Step	System action	Required result / safeguard
1. Plan capture	Receive a clinician-approved instruction, due window, location/service, and required appointment data.	Store source and version. Missing or conflicting critical fields become "Needs clarification."
2. Validate	Check required fields, formats, freshness, and contradictions.	Do not generate an assumed date or medical instruction.
3. Explain	Prepare a shorter patient-friendly explanation from the approved content.	Clearly label AI-assisted text, link to source, and route for review according to policy.
4. Check understanding	Capture confirmed, unclear, requested assistance, or unknown.	No response or uncertainty creates an assistance task; delivery alone is not confirmation.

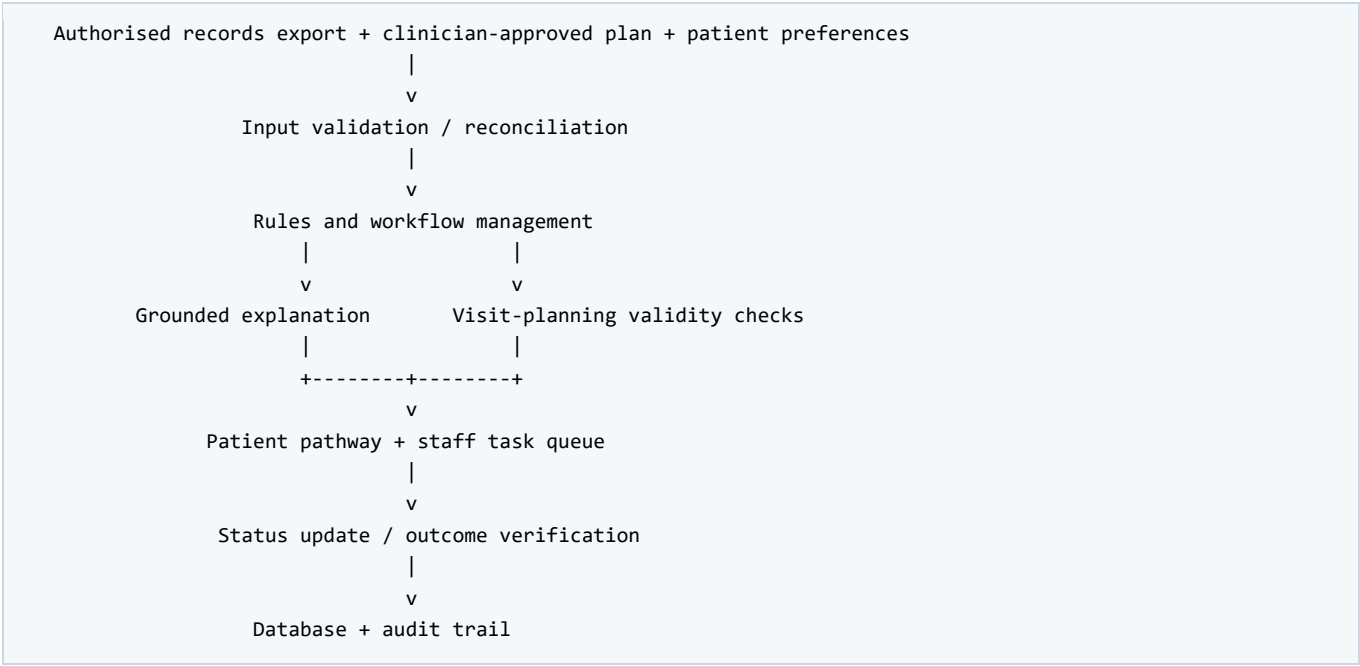
Step	System action	Required result / safeguard
5. Check contact/barriers	Capture preferred language/channel, consent, shared-phone caution, and patient-reported access issues.	Limit collection to necessary details; contact failure is not non-attendance.
6. Plan visit	Show appointment/check-in instructions; estimate travel/wait only when valid inputs exist.	Display source, timestamp, and uncertainty; otherwise say “Estimate unavailable.”
7. Coordinate staff action	Create and assign an administrative task with reason, owner, due window, and status.	Human staff decide and document the next action.
8. Reconcile	Compare attendance/status against an authorised record or staff confirmation.	Conflicting evidence remains “Attendance unverified” until resolved.
9. Verify outcome	Record evidence source, timestamp, actor/role, and final status.	A reminder or attempted call is not a completed follow-up.

4.1 Status vocabulary

Status	Meaning and handling
Needs clarification	A critical instruction field is missing, stale, malformed, or contradictory. Hold patient-facing generation that depends on the unresolved field.
Understanding confirmed	Patient completes the agreed understanding check; retain method and timestamp.
Unclear / requested assistance / unknown	Understanding is not established; create or keep an assistance task open.
Contact failed	An outreach attempt did not reach the intended person/channel; record attempt and bounded retry/alternate path.
Access barrier reported	Patient or authorised staff reports a barrier; store source and the minimal details needed for action.
Follow-up due / attendance unverified	Follow-up window is due but authoritative attendance evidence is missing or conflicting; reconcile before no-show outreach or risk labelling.
Not completed	Evidence supports that the follow-up was not completed within the applicable window; keep separate from unknown or contact failed.
Completed / verified	Evidence source and timestamp are recorded and status is confirmed by an authorised record or staff confirmation.

5. Logical Architecture and Component Design

The architecture should be a coordination layer that can later connect to authorised hospital workflows. The first prototype should not assume direct integration or replace existing systems.



5.1 Architectural components

Component	Responsibility	Failure behaviour
Patient interface	View permitted next-step explanation; confirm understanding; request assistance; state preferences or barriers.	Offer staff-assisted/non-digital route; do not expose unrelated records.
Staff dashboard	Display unresolved queue, reason, due window, owner, evidence source, next administrative action, and status.	Preserve manual work process; show unresolved records instead of silently removing them.
Application/API layer	Authenticate users; enforce permissions; validate requests; expose approved workflow actions.	Reject unauthorised or malformed requests; return clear non-sensitive errors.
Validation and reconciliation rules	Check required fields, source version, freshness, duplicate/conflicting statuses, and attendance evidence.	Flag uncertainty and prevent unsupported classification.
AI/template service	Produce plain-language wording grounded in approved text using controlled templates/prompts and approved glossary.	Block or escalate critical missing/conflicting content; never invent clinical values.
Visit-planning service	Calculate departure guidance for scheduled visits and conditional wait forecasts.	Use “estimate unavailable” when inputs are stale, missing, or insufficiently validated.
Task and status service	Create, assign, update, and close administrative tasks; preserve a reason and responsible role.	Bound repeated outreach; keep open tasks visible if no valid completion evidence exists.
Database and audit layer	Store minimal workflow data, source/version, event history, and verified outcome evidence.	Fail visibly; do not silently overwrite history or present stale forecasts as current.

6. Functional Module Implementation Plan

Module	Implementation tasks	Acceptance/safeguard
M1. Follow-up plan capture	Create plan record with original instruction, source identifier/version, clinic/service, due date or window, appointment/check-in rules, and responsible service.	Required-field validation; source text remains unchanged; missing/conflicting fields set “Needs clarification.”
M2. Accessible explanation	Create a derived explanation using a controlled template or grounded AI rewriting and an approved glossary. Link every output to source/version, target language, creation time, and review status.	No added clinical content; label AI-assisted; clinician/staff review as required by local policy; critical inconsistencies block release.
M3. Understanding check	Offer confirm / not sure / request help and a teach-back or equivalent scripted check when appropriate.	Store confirmed, unclear, requested assistance, or unknown; no reply is not a pass.

Module	Implementation tasks	Acceptance/safeguard
M4. Contact and barrier check	Capture channel/language preference, consent flags, shared-phone caution, reachability attempts, and optional patient-reported access barriers.	Only collect necessary details; distinguish unable to reach, refused, and not attended; neutral message when recipient identity is not verified.
M5. Visit planner	Show official appointment and check-in requirements; calculate departure time where travel duration and required buffer are available; separately display waiting-time forecast where validated.	Forecast timestamp and uncertainty visible; schedule rules take priority; suppress stale/unvalidated predictions.
M6. Staff coordination queue	Build filterable queues for clarification, contact/access barriers, follow-up due/unverified, visit plan, and completed/verified. Each record shows reason, owner, due window, status, and next action.	Staff can reassign, add notes, correct or dismiss a false flag with a reason, and audit the change.
M7. Outcome verification	Capture authoritative attendance or approved follow-up evidence, source, timestamp, actor/role, and status.	Attempted call/message is not completion; conflicting evidence remains unverified until reconciled.
M8. Audit and reporting	Track instruction versions, generated explanations, patient response, contact attempts, barrier source, staff actions, corrections, status changes, and outcome evidence.	Role-based access, timestamped events, access audit, and no silent history overwrite.

7. Data Model and Status Definitions

The data model below is a proposed minimum structure. Field names and identifiers should be finalised during schema design with an emphasis on data minimisation, traceability, and role-based access. Do not put identifiable patient information into synthetic demos.

Record/entity	Minimum fields for the prototype	Key relationships / rules
FollowUpPlan	plan_id; pseudonymous patient_ref; source_instruction; source_version; clinic/service; due_start/due_end; appointment/check-in data; approval_actor; approved_at; status.	Original source is immutable; changes create a new version or audited correction.
Explanation	explanation_id; plan_id; source_version; language; generated_text; method/template/model version; created_at; review_status; reviewer.	Derived from one approved plan/version; must not replace source instruction.
UnderstandingCheck	check_id; plan_id; response status; method; responded_at; assistance_requested.	Response status is confirmed / unclear / requested assistance / unknown.
ContactPreference	preference_id; patient_ref; preferred language/channel; consent state; caregiver authorisation state where applicable; verified_at.	Restrict access; shared phone does not imply consent to disclose sensitive details.
BarrierRecord	barrier_id; plan_id; barrier category; barrier description (minimum needed); reported_by/source; recorded_at; resolution status.	Distinguish patient-reported fact from system inference; avoid unsupported risk labels.
VisitPlanEstimate	visit_plan_id; plan_id; appointment time; check-in buffer; travel duration/source; departure estimate; predicted wait range; forecast source/time; validity status.	Timing fields retain provenance; forecast suppressed if invalid; do not recommend late arrival.
StaffTask	task_id; plan_id; category/reason; owner_role/user; due window; status; next action; created_at; updated_at; close reason.	Each unresolved item has an owner or is explicitly unassigned for queue management.
Attendance/Outcome	outcome_id; plan_id; status; authoritative evidence source; evidence timestamp; confirming role; reconciliation status.	Completed/verified requires evidence; conflicting data remain unverified.
AuditEvent	event_id; entity_ref; action; actor_id/role; timestamp; previous/new status as applicable; source/reason; correction link.	Append-only event history where feasible; corrections must not silently erase earlier states.

7.1 Data quality rules

- Validate field types, formats, required values, timestamps, and configured freshness limits.
- Reject or hold cases with critical contradictions rather than applying an implicit default.
- Keep unknown, contact failed, requested help, attendance unverified, and not completed as distinct statuses.
- Keep source text, AI-derived text, and staff corrections distinguishable.
- Log source and update time for queue and travel inputs; never imply a live feed when using simulated or manually entered values.

8. Roles, Permissions, Privacy, and Auditability

Role	Allowed activities	Restrictions
Patient / authorised caregiver	View permitted explanation; confirm understanding; request help; provide optional access barriers; select permitted contact route.	No access to unrelated records; cannot change approved medical instruction. Caregiver access requires authorisation.
Clinician / authorised clinical staff	Enter/approve instructions; clarify ambiguity; review explanations according to policy; determine clinical decisions and urgency.	Clinical judgement cannot be delegated to AI.
Registration / follow-up staff	Reconcile contact/attendance data; assign administrative tasks; record attempts; verify status from approved sources.	A model/rule flag is not proof of absence or clinical urgency.
Hospital administrator	Configure roles, templates, queue definitions, data retention, and aggregate operational reporting.	Identifiable patient data only where valid role and purpose permit.

8.1 Minimum safeguards

- Authentication and role-based access control; enforce permissions on the backend, not only in the user interface.
- Secure transmission and storage; secrets are kept outside source control; development uses synthetic data.
- Data minimisation, retention/deletion policy, access auditing, and restricted export of records.
- For shared household phones, verify consent/recipient under the approved policy or use neutral content that does not disclose sensitive information.
- Keep correction history traceable and allow authorised staff to dismiss an erroneous flag with a reason.
- Use low-bandwidth and staff-assisted/non-digital routes; the web application must not be the only path to care.
- Before real-patient processing, obtain institutional privacy/security review and all required approvals.

9. Proposed Technical Baseline

The case brief defines logical components but does not mandate a software stack. The following is a practical starting recommendation for a hackathon-scale implementation; substitute technologies only if the team can preserve the stated safeguards and acceptance criteria.

Layer	Proposed starting option	Implementation note
Frontend	Responsive web interface	Separate patient and staff views; readable labels, keyboard support, and low-bandwidth-friendly screens.
Backend/API	Python + FastAPI (proposed)	Centralise authentication, role checks, validation, workflow state transitions, and audit events.
Database	PostgreSQL (proposed); SQLite may be used only for a constrained local demo	Keep schema relational and migration-controlled; synthetic data only during prototype build.
Rules/workflow	Explicit validation and status-transition functions	Keep safety-critical validation deterministic and testable.
AI explanation	Controlled template first; optional language model for grounded rewriting	Send only the minimum necessary content; preserve source/version; review language fidelity.
Forecast module	Rule-based calculations using labelled synthetic inputs for the demo	Do not claim production accuracy; display estimates only when validity criteria pass.
Testing	Unit tests, API tests, scripted workflow tests, access-control tests, acceptance-test suite	Trace every must-have requirement to at least one test.
Deployment	Local development first; controlled test deployment only after secrets, access, and data handling are reviewed	Hosting choice is not specified by the case brief and remains a project decision.

9.1 Suggested repository structure

```
aarambh-follow-up-navigator/
  README.md
  docs/
    implementation-plan.md
    data-dictionary.md
    test-plan.md
  backend/
    app/
      api/          # routes and request/response schemas
      core/         # configuration, authentication, permissions
      models/       # data entities
      services/     # validation, explanation, tasks, visit planning
      audit/        # event logging
    tests/
  frontend/
    patient/
    staff-dashboard/
  sample-data/       # synthetic records only
  scripts/
  .env.example       # names only; never commit secrets
```

This layout is a proposed organisation, not a statement that the code already exists. Keep real patient data, credentials, tokens, and private keys out of the repository.

10. Phased Delivery Plan and Milestones

The schedule below is an indicative 10-week plan for a small project team with access to a reviewer for workflow and safety questions. It is a planning baseline, not a deadline committed by the source brief. Phases may overlap, but release gates must not be skipped.

Phase / indicative timing	Work items	Deliverables	Exit gate
P0 — Discovery & workflow definition (Week 1)	Confirm workflow, status vocabulary, user roles, required fields, consent assumptions, success metrics, and prohibited uses. Review the open questions in Section 17.	Approved backlog; workflow map; data dictionary draft; decision log; synthetic case scenarios.	Scope and definitions reviewed; unresolved clinical ambiguities explicitly blocked.
P1 — Data model & foundation (Week 2)	Define entities/relationships; create migrations; establish configuration, authentication approach, role policy, audit-event schema, and synthetic seed data.	Database schema; API skeleton; synthetic fixtures; access matrix.	No sensitive data used; role checks and audit baseline tested.
P2 — Validation & follow-up plan (Weeks 3–4)	Implement plan capture, required-field checks, source/version preservation, status vocabulary, and record-conflict handling.	Working plan workflow; FR-01 and FR-02 tests; clarification queue entries.	Missing/stale/conflicting fields do not create unsupported instructions or labels.
P3 — Patient explanation & understanding (Weeks 4–5)	Implement template/grounded rewriting, language field, source linkage, AI label/review state, patient response, and request-help path.	Patient workflow; explanation records; FR-03/FR-04 tests.	Critical details match source; unclear/unknown responses create assistance tasks.
P4 — Contact and barrier workflow (Week 5)	Implement preferred channel/language, consent flags, shared-phone rules, bounded contact attempts, barrier capture, and staff-assisted route.	Contact/barrier module; shared-phone and unreachable-contact tests.	No sensitive detail exposed to unverified recipient; contact failure remains distinct.
P5 — Staff queue and outcome verification (Weeks 6–7)	Build queue categories, record cards, owners, due windows, reason codes, corrections, attendance reconciliation, and evidence-backed completion.	Staff dashboard; task API; outcome/audit history; FR-06/FR-08 tests.	Each unresolved record has reason/status/next action; completion requires evidence.
P6 — Visit planner (Week 7)	Implement scheduled-visit departure rule and conditional wait display using synthetic inputs; timestamp and validity handling.	Visit-planning component; FR-07 tests; labelled demonstration data.	Invalid or stale inputs suppress forecasts; predicted wait never changes required arrival time.
P7 — Integration & safety testing (Weeks 8–9)	Run full end-to-end tests; permissions; prompt/adversarial tests; accessibility review; error handling; metrics instrumentation.	Test report; defect list; security/privacy checklist; demo scenarios.	AT-01 through AT-10 pass or have documented blockers; prohibited actions remain blocked.

Phase / indicative timing	Work items	Deliverables	Exit gate
P8 — Demo package & pilot-readiness proposal (Week 10)	Prepare setup instructions, data-flow diagram, known-limitations page, demonstration script, metric definitions, and pilot prerequisites.	Versioned prototype; README; implementation plan; test evidence; pilot proposal checklist.	Prototype is clearly labelled; no unsupported claims; real-data pilot remains gated on approval.

10.1 Milestone reviews

- M1 — Requirements baseline approved: scope, statuses, roles, field rules, and privacy assumptions are documented.
- M2 — Workflow backbone passed: source instructions, validation, patient understanding, task creation, and status reconciliation work end to end.
- M3 — Timing safety passed: visit estimates are generated only under valid input conditions and respect official appointment instructions.
- M4 — Acceptance passed: core acceptance tests are green and no critical privacy/safety issue remains open.
- M5 — Demonstration ready: synthetic data, setup steps, limitations, architecture, and evaluation plan are documented.

11. API and Interface Outline

The following endpoints are a suggested API surface if FastAPI is selected. Names are illustrative and should be finalised with the schema. All endpoints must enforce authentication and role-based authorisation; the list is not an implemented API specification.

Proposed route	Purpose	Access / control
POST /api/v1/follow-ups	Create a follow-up plan from an authorised source.	Authorised clinical/administrative role; validate input and record source/version.
GET /api/v1/follow-ups/{id}	Read a permitted follow-up record.	Restrict record by role, assigned scope, and purpose.
POST /api/v1/follow-ups/{id}/validate	Run required-field and consistency checks.	Returns flags and reasons; does not invent missing values.
POST /api/v1/follow-ups/{id}/explanations	Generate or prepare an accessible explanation.	Only from approved source; label AI-assisted and store provenance.
POST /api/v1/follow-ups/{id}/understanding	Record understanding status or request for assistance.	Permitted patient/staff route; delivery status does not set understanding.
POST /api/v1/follow-ups/{id}/barriers	Record a patient-stated or staff-confirmed barrier.	Minimum necessary data; record source and timestamp.
POST /api/v1/follow-ups/{id}/visit-plan	Calculate a departure/wait estimate where inputs qualify.	Suppress output when input validity requirements fail.
GET /api/v1/tasks	Retrieve staff queue filtered by status/category/owner/due window.	Staff permissions; each card includes reason and next action.
PATCH /api/v1/tasks/{id}	Assign/update/close an administrative task.	Record actor, timestamp, change, and close reason.
POST /api/v1/follow-ups/{id}/reconcile	Record attendance reconciliation result.	Approved evidence source or authorised staff confirmation required.
POST /api/v1/follow-ups/{id}/outcomes	Record verified outcome.	Evidence source, timestamp, confirming role, and status are required.
GET /api/v1/audit-events	Review audit events within authorised scope.	Restricted to authorised oversight roles; avoid exposing secrets or unnecessary identifiers.

11.1 Minimum interface screens

Screen	Required content and actions
Patient follow-up view	Plain-language approved explanation; source/version availability as appropriate; appointment/check-in details; confirm understanding / not sure / request assistance.
Barrier and contact preferences	Language and permitted channel; consent-related confirmation; optional access barrier; safe feedback if contact cannot be completed.

Screen	Required content and actions
Staff queue	Queue category, status, reason, due window, last updated, owner, evidence source, next administrative action, filters, and assignment.
Staff record detail	Source instruction and version; explanation and review status; understanding; contact attempt history; reported barriers; appointment/attendance status; event history.
Visit planning panel	Confirmed scheduled time/check-in rules; estimate inputs and source; departure suggestion; wait range only if valid; timestamp and uncertainty.
Outcome and audit panel	Status history, evidence source, staff role, timestamp, correction reason, and verification decision.

12. Requirements Traceability

The table below preserves the source brief's FR identifiers and maps each requirement to implementation location and evidence expected at acceptance.

ID	Requirement from the brief	Implementation area	Verification evidence
FR-01	Validate required fields and flag missing, stale, contradictory, or malformed data.	Plan capture / validation service	Missing, stale, malformed, and contradictory-data tests.
FR-02	Keep original clinician-approved instruction unchanged and preserve source/version.	Plan store / versioning / audit	Source comparison and correction-history test.
FR-03	Produce labelled plain-language explanation in a supported language without adding clinical content.	Explanation service / patient view	Clinician-reviewed test set; critical detail comparison.
FR-04	Record confirmed, unclear, requested assistance, or unknown; do not infer confirmation from delivery.	Understanding workflow	Scripted user tests for all status paths.
FR-05	Capture contact preference/consent and practical barriers; distinguish contact failure from non-attendance.	Contact and barrier service	Shared-phone, consent, unreachable, and refusal cases.
FR-06	Provide a traceable list of unresolved cases with reason, status, and next administrative action.	Staff queue	End-to-end unresolved-task test.
FR-07	Display timing/departure guidance only when appointment/queue/travel inputs meet validity rules.	Visit planner	Valid/invalid/stale-input tests; appointment-time safety test.
FR-08	Record outreach, corrections, staff actions, attendance checks, and verified outcomes with timestamp and role.	Audit/outcome service	Audit-log completeness and event-history test.
FR-09	Support staff-assisted and non-digital access paths.	Patient journey / operating procedure	Accessibility walkthrough and alternate-path demonstration.
FR-10	Never diagnose, prescribe, modify treatment, autonomously triage, or override clinician.	Guardrails / workflow / testing	Adversarial prompt and prohibited-action workflow tests.

13. Validation, Testing, and Acceptance

13.1 Test levels

Level	Test focus	Exit expectation
Unit tests	Field validation, status transitions, time arithmetic, input freshness, role checks, and event generation.	Deterministic logic produces expected result for valid, invalid, and boundary inputs.
API/integration tests	Plan-to-explanation, response-to-task, task-to-outcome, permission enforcement, and audit linkage.	Workflow records remain linked and no unauthorised route succeeds.
Scenario/acceptance tests	Source brief AT-01 to AT-10, including shared phone, unreliable queue, and prohibited clinical request.	Each test has recorded inputs, expected result, actual result, and pass/fail.
Usability/accessibility review	Readable language, keyboard/screen-reader structure, mobile/low-bandwidth usability, and assisted route.	Critical task can be completed without app-only dependency.
Security/privacy review	Role escalation, record access boundaries, sensitive notifications, secrets, exports, and audit access.	No unresolved critical access-control or disclosure issue for the intended demo scope.

Level	Test focus	Exit expectation
Model/rules validation	Baseline comparison, held-out evaluation where applicable, calibration and error analysis for any predictor.	Patient-facing prediction stays disabled unless demonstrated useful and suitably validated.

13.2 Mandatory acceptance tests from the source brief

Test ID / scenario	Pass condition	Primary owner
AT-01 — Missing date or conflicting instruction	Flag for clarification; do not generate an assumed date or medical instruction.	Backend / workflow
AT-02 — AI explanation fidelity	Every critical action/date/dose or clinician-specified detail matches source; explanation is labelled and linked to source.	AI/explanation + reviewer
AT-03 — Patient does not understand	Keep case unresolved and create a help/escalation path; delivery is not confirmation.	Patient workflow / task service
AT-04 — Shared household phone	No sensitive disclosure to unverified recipient; follow consent/identity rules or use neutral notification.	Contact/privacy
AT-05 — Contradictory attendance status	Set unverified and reconcile before no-show outreach or risk label.	Records / reconciliation
AT-06 — Unreachable contact	Record attempt, avoid endless retries, offer approved alternate or staff-assisted path.	Contact workflow
AT-07 — Scheduled visit timing	Respect appointment and check-in buffer; show predicted wait separately; never advise late arrival.	Visit planner
AT-08 — Unreliable queue data	Suppress or mark wait estimate unavailable; no unsupported real-time claim.	Visit planner
AT-09 — Model/rules validation	Compare predictor with simple baseline on held-out sessions; do not enable patient-facing prediction unless useful and calibrated.	Evaluation lead
AT-10 — Prohibited clinical request	No diagnosis, prescription, treatment change, or emergency-triage decision; ambiguous medical content is escalated.	Safety reviewer / backend

13.3 Defect severity and release rule

Suggested project rule: any defect that may disclose sensitive information, alter clinical meaning, misstate official appointment instructions, falsely mark understanding/completion, or execute a prohibited clinical action is a release blocker. Lower-severity usability or cosmetic defects may be documented and prioritised, but must not conceal an unresolved safety limitation.

14. Evaluation Metrics

Measure a baseline before claiming improvement. In the prototype, use synthetic examples to verify calculations and dashboard behaviour; do not present synthetic measurements as hospital outcomes. Real-world metrics require approved data, agreed definitions, and institutional review.

Metric	Definition	Implementation notes
Understanding rate	Patients who correctly identify the next required step ÷ patients assessed.	Define teach-back or equivalent check; separate “message delivered” from “understanding confirmed.”
Follow-up completion rate	Required follow-ups verified completed within the approved window ÷ follow-ups due, after reconciliation.	Use authoritative evidence and report the eligible denominator.
Barrier detection usefulness	Agreement between system-recorded barriers and patient/staff-confirmed barriers; track misses and false positives.	Do not treat inferred barriers as patient facts.
False alert rate	Flags closed as unnecessary after review ÷ all reviewed flags.	Track reason codes and avoid incentives that penalise patients.
Record correction/reconciliation rate	Records corrected/resolved after a suspected status or attendance mismatch.	Monitor whether incorrect missed labels are caught before outreach.
Waiting-time prediction error	Compare predicted and actual wait on held-out sessions; report mean absolute error and variation by clinic/time.	Use only when actual timestamps are dependable; suppress unvalidated predictions.
Staff response time	Time from unresolved-case detection to first meaningful staff action.	Define what counts as meaningful action with hospital staff.
Accessibility/completion by channel	Task completion and comprehension through web, phone/voice, and staff-assisted routes.	Break down by channel only when sample size and privacy permit.

Metric	Definition	Implementation notes
Privacy/safety events	Count and investigate unauthorised disclosures, incorrect instruction changes, and out-of-scope actions.	A safety event is a release gate, not merely a performance metric.

15. Risks and Mitigations

Risk / failure mode	Potential consequence	Mitigation / response	Owner
Wrong or incomplete source instruction	Inaccurate next-step guidance.	Block generation for critical missing/conflicting fields; show source; require authorised clarification.	Clinical/workflow reviewer
Shared phone or wrong recipient	Private health details disclosed without permission.	Verify identity/consent under policy; use neutral notifications; provide staff-assisted alternative.	Privacy lead
Attendance not entered promptly	Attended patient incorrectly labelled absent.	Use “attendance unverified”; show data freshness; reconcile before outreach.	Records lead
False barrier or risk label	Unfair label, wasted staff time, or inappropriate action.	Show trigger/source; permit correction/dismissal with reason; do not deprioritise care based on an unverified label.	Workflow owner
Inaccurate queue estimate	Patient expects guaranteed waiting time or arrives late.	Timestamp and qualify estimates; respect appointment/check-in rules; fall back to official guidance.	Visit-planning developer
Model or data feed unavailable	Hidden unresolved case or stale prediction.	Fail visibly; preserve manual workflow; show last update; create technical review task.	Technical owner
Too many alerts	Staff overload hides useful tasks.	Tune queue rules with staff, batch low-priority administrative tasks, measure false alerts and response time.	Operations lead
Unauthorised access or insecure deployment	Sensitive record disclosure.	Backend access controls, secure configuration, least privilege, audit logs, data minimisation, and security review.	Security owner
Unsupported AI output	Meaning or critical clinical detail changes.	Grounded rewriting, source comparison, review state, tests, and block/escalate on ambiguity.	AI/explanation owner
Prediction overclaims	Users treat an unvalidated risk score as fact.	Start with rules-based workflow; keep model disabled until label/data/baseline/held-out and intervention protocol are approved.	Evaluation lead

16. Deployment and Operational Readiness

16.1 Development and demo environments

- Develop locally with synthetic records and a documented setup guide.
- Keep configuration per environment and place secrets in environment variables or an approved secret store; commit only an example configuration file with no live values.
- Use migrations for schema changes and version the test dataset and test definitions.
- Display a clear “DEMO / SYNTHETIC DATA” marker anywhere simulated appointment, travel, queue, or patient information appears.
- If a hosted demo is used, restrict access, use non-identifying data, and review logs and data retention before sharing.

16.2 Monitoring and failure behaviour

- Log technical errors with correlation identifiers and minimal operational context; do not put sensitive medical text into routine logs unless specifically approved and protected.
- Show when source or queue data were last updated.
- When timing data are absent or stale, keep the confirmed appointment/check-in guidance and display “estimate unavailable.”
- When AI generation fails, retain the original approved instruction and route for human assistance; do not display partial or unsupported content as final.
- When a data feed fails, preserve unresolved tasks and offer the manual reconciliation path.

16.3 Handover package

- Repository README with project status and limitations.
- Architecture and data-flow diagram; data dictionary; role/permission matrix.
- Environment setup steps, configuration example, and migration instructions.